Comparison of Subspace Convergence Rates.

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Proof. Let $X \in \mathbb{R}^d$ be a random input vector. Define the EGOP matrix

$$H \;:=\; \mathbb{E}\big[\nabla f(X)\,\nabla f(X)^\top\big] \in \mathbb{R}^{d\times d}.$$

Given i.i.d. samples $X_1, \dots, X_n \sim X$, define the random matrices

$$Z_i \;:=\; \nabla f(X_i)\,\nabla f(X_i)^\top \in \mathbb{R}^{d\times d}, \qquad H_n \;:=\; \frac{1}{n}\sum_{i=1}^{n} Z_i.$$

Goal: control the estimation error $\|H_n - H\|$.

Verify matrix Bernstein conditions.

1. **Independence.** The Z_i are independent since the X_i are i.i.d.
2. **Mean.** For each i,

$$\mathbb{E}[Z_i] = \mathbb{E}\big[\nabla f(X_i)\nabla f(X_i)^\top\big] = \mathbb{E}\big[\nabla f(X)\nabla f(X)^\top\big] = H.$$

3. **Boundedness of each term.** Assume (From Assumption 3 in the paper) that

$$\|\nabla f(x)\|_2 \le L \quad \text{for all } x.$$

   Since for a rank-one matrix $vv^\top$ we have $\|vv^\top\| = \|v\|_2^2$, it follows that

$$\|Z_i\| = \big\|\nabla f(X_i)\nabla f(X_i)^\top\big\| = \|\nabla f(X_i)\|_2^2 \le L^2.$$

Setup Define

$$Y_i \;:=\; Z_i - \mathbb{E}[Z_i] \;=\; Z_i - H.$$

Then $\mathbb{E}[Y_i] = 0$, the Y_i are independent, and by triangle inequality

$$\|Y_i\| \le \|Z_i\| + \|H\|.$$

Also, since $H = \mathbb{E}[Z_i]$ and $\|Z_i\| \le L^2$, Jensen yields $\|H\| \le \mathbb{E}\|Z_i\| \le L^2$. Hence

$$\|Y_i\| \le L^2 + L^2 = 2L^2.$$

So we may apply the matrix Bernstein inequality to

$$H_n - H = \frac{1}{n}\sum_{i=1}^{n} Y_i,$$

and therefore we need to bound

$$\left\|\frac{1}{n}\sum_{i=1}^{n} Y_i\right\|.$$

Variance proxy. Set

$$\sigma^2 := \left\| \sum_{i=1}^{n} \mathbb{E}\big[Y_i^2\big] \right\|.$$

Each Y_i is symmetric because both $Z_i = \nabla f(X_i)\nabla f(X_i)^\top$ and H are symmetric, and the difference of symmetric matrices is symmetric. For symmetric matrices A, $\|A^2\| = \|A\|^2$, so

$$\|Y_i^2\| = \|Y_i\|^2 \le (2L^2)^2 = 4L^4.$$

Then

$$\big\|\mathbb{E}[Y_i^2]\big\| \le \mathbb{E}\|Y_i^2\| \le 4L^4,$$

and therefore

$$\sigma^2 = \left\| \sum_{i=1}^{n} \mathbb{E}[Y_i^2] \right\| \le \sum_{i=1}^{n} \big\|\mathbb{E}[Y_i^2]\big\| \le 4nL^4.$$

Apply matrix Bernstein. Matrix Bernstein states that if $\|Y_i\| \le R$ and $\sigma^2 = \big\|\sum_{i=1}^{n} \mathbb{E}[Y_i^2]\big\|$, then for all $t \ge 0$,

$$\mathbb{P}\left(\left\| \sum_{i=1}^{n} Y_i \right\| \ge t \right) \le 2d \exp\left(-\frac{t^2/2}{\sigma^2 + Rt/3} \right).$$

Let $S := \sum_{i=1}^{n} Y_i$. Since $H_n - H = \frac{1}{n} S$,

$$\mathbb{P}\big(\|H_n - H\| \ge \varepsilon\big) = \mathbb{P}\big(\|S\| \ge n\varepsilon\big) \le 2d \exp\left(-\frac{(n\varepsilon)^2/2}{4nL^4 + (2L^2)(n\varepsilon)/3} \right).$$

Equivalently,

$$\mathbb{P}\big(\|H_n - H\| \ge \varepsilon\big) \le 2d \exp\left(-\frac{n\varepsilon^2}{2\left(4L^4 + \frac{2}{3}L^2\varepsilon\right)} \right).$$

Simplification.

1. **Case 1:** $\varepsilon \le 6L^2$**.** Then $\frac{2}{3}L^2\varepsilon \le \frac{2}{3}L^2 \cdot 6L^2 = 4L^4$, so

   $$4L^4 + \frac{2}{3}L^2\varepsilon \le 8L^4,$$

   hence

   $$\mathbb{P}\big(\|H_n - H\| \ge \varepsilon\big) \le 2d \exp\left(-\frac{n\varepsilon^2}{16L^4} \right) = 2d \exp\left(-c\,\frac{n\varepsilon^2}{L^4} \right),$$

   where one can take $c = \frac{1}{16}$.

2. **Case 2:** $\varepsilon > 6L^2$**.** Then $\frac{2}{3}L^2\varepsilon \ge 4L^4$, so

   $$4L^4 + \frac{2}{3}L^2\varepsilon \le \frac{4}{3}L^2\varepsilon + \frac{2}{3}L^2\varepsilon = 2L^2\varepsilon,$$

   giving

   $$\mathbb{P}\big(\|H_n - H\| \ge \varepsilon\big) \le 2d \exp\left(-\frac{n\varepsilon^2}{4L^2\varepsilon} \right) = 2d \exp\left(-\frac{n\varepsilon}{4L^2} \right).$$

   This decays faster in ε and does not affect the final (sub-Gaussian) rate. So we will focus on results from Case 1 later.

High-probability bound. Set the Case-1 RHS equal to δ:

$$2d\exp\left(-c\,\frac{n\varepsilon^2}{L^4}\right) \le \delta \quad \Longrightarrow \quad \varepsilon \;\approx\; L^2\sqrt{\frac{\log d + \log(1/\delta)}{n}}.$$

Thus with probability at least $1-\delta$,

$$\|H_n - H\| \;\lesssim\; L^2\sqrt{\frac{\log d + \log(1/\delta)}{n}}.$$

Expectation bound. Let $Z := \|H_n - H\|$. Using $\mathbb{E}[Z] = \int_0^\infty \mathbb{P}(Z \ge t)\,dt$ and the tail bound

$$\mathbb{P}(Z \ge t) \le \min\left\{2d\exp\big(-c\,\tfrac{nt^2}{L^4}\big),\, 1\right\},$$

choose t_0 such that $2d\exp(-c\,nt_0^2/L^4) = 1$, i.e.

$$t_0 = \frac{L^2}{\sqrt{cn}}\sqrt{\log(2d)}.$$

Then

$$\mathbb{E}[Z] \le \int_0^{t_0} 1\,dt + \int_{t_0}^\infty 2d\exp\left(-c\,\frac{nt^2}{L^4}\right)\,dt \;\lesssim\; L^2\sqrt{\frac{\log d}{n}}.$$

(The complete calculation for the above equation is at the end of the document).

Davis–Kahan subspace error. Let $U \in \mathbb{R}^{d\times r}$ contain the top r eigenvectors of H, and suppose there is an eigengap

$$\delta := \lambda_r(H) - \lambda_{r+1}(H) > 0.$$

Let U_n contain the top r eigenvectors of H_n. Davis–Kahan gives

$$\|\sin\Theta(U_n, U)\| \le \frac{\|H_n - H\|}{\delta}.$$

Taking expectations,

$$\mathbb{E}\|\sin\Theta(U_n, U)\| \le \frac{1}{\delta}\,\mathbb{E}\|H_n - H\| \;\lesssim\; \frac{L^2}{\delta}\sqrt{\frac{\log d}{n}}.$$

Comparing gradient vs. without-gradient rates. For the gradient version, the subspace error bound behaves like

$$\mathcal{O}\left(\frac{L^2}{\delta}\sqrt{\frac{\log d}{n}}\right).$$

For the without-gradient version (as written in the paper),

$$\mathbb{E}\|\sin\Theta(\widehat{U}_n, U)\| \le \frac{1}{\delta}\,n^{-\frac{3}{4d+12}}, \qquad \text{so subspace rate } \mathcal{O}\left(\frac{1}{\delta}\,n^{-\frac{3}{4d+12}}\right).$$

Since for $d \ge 1$ we have $\frac{3}{4d+12} < \frac{1}{2}$, the without-gradient rate is slower. Indeed,

$$\frac{\sqrt{\log d}\,n^{-1/2}}{n^{-3/(4d+12)}} = \sqrt{\log d}\;n^{-\left(\frac{1}{2}-\frac{3}{4d+12}\right)} \to 0 \quad (d \ge 1),$$

so the gradient version's subspace error upper bound approaches 0 faster. In summary, incorporating gradient information yields a faster rate of convergence for subspace estimation. Therefore, the gradient-based method achieves strictly faster subspace recovery compared to the without-gradient approach. □

Appendix: Calculation for $\mathbb{E}\|H_n - H\|$

Let

$$H_n - H = \frac{1}{n}\sum_{i=1}^{n} Y_i, \qquad Z := \|H_n - H\|.$$

Tail bound (from matrix Bernstein)

From the previous steps, we have

$$\mathbb{P}(\|H_n - H\| \geq \varepsilon) \leq 2d \exp\left(-\, c\, \frac{n\varepsilon^2}{L^4}\right)$$

Using $\mathbb{E}[Z] = \int_0^\infty \mathbb{P}(Z \geq t)\, dt$ and the fact that probability cannot exceed 1, we have

$$\mathbb{P}(Z \geq t) \leq \min\left\{ 2d \exp\left(-c\, \frac{nt^2}{L^4}\right),\, 1 \right\}.$$

Expectation as an integral and splitting at t_0

Use

$$\mathbb{E}[Z] = \int_0^\infty \mathbb{P}(Z \geq t)\, dt \leq \int_0^\infty \min\left\{ 2de^{-cnt^2/L^4},\, 1 \right\} dt.$$

Choose t_0 such that

$$2d \exp\left(-c\, \frac{nt_0^2}{L^4}\right) = 1.$$

Then

$$\exp\left(-c\, \frac{nt_0^2}{L^4}\right) = \frac{1}{2d} \quad \Longrightarrow \quad c\, \frac{nt_0^2}{L^4} = \ln(2d) \quad \Longrightarrow \quad t_0 = \frac{L^2}{\sqrt{cn}} \sqrt{\ln(2d)}.$$

Hence

$$\mathbb{E}[Z] \leq \int_0^{t_0} 1\, dt + \int_{t_0}^\infty 2d \exp\left(-c\, \frac{nt^2}{L^4}\right) dt = t_0 + \int_{t_0}^\infty 2d \exp\left(-c\, \frac{nt^2}{L^4}\right) dt.$$

Bounding the tail integral

Let

$$u := \sqrt{cn}\, \frac{t}{L^2}, \qquad \text{so} \qquad t = \frac{L^2}{\sqrt{cn}} u, \qquad dt = \frac{L^2}{\sqrt{cn}}\, du.$$

When $t = t_0$, we have

$$u_0 = \sqrt{cn}\, \frac{t_0}{L^2} = \sqrt{\ln(2d)}.$$

Therefore,

$$\int_{t_0}^\infty 2d \exp\left(-c\, \frac{nt^2}{L^4}\right) dt = 2d \cdot \frac{L^2}{\sqrt{cn}} \int_{u_0}^\infty e^{-u^2}\, du.$$

Using the standard bound,

$$\int_{u_0}^\infty e^{-u^2}\, du \leq \frac{1}{2u_0} e^{-u_0^2} \qquad \text{for } u_0 > 0,$$

we get

$$\int_{t_0}^\infty 2d \exp\left(-c\, \frac{nt^2}{L^4}\right) dt \leq 2d \cdot \frac{L^2}{\sqrt{cn}} \cdot \frac{1}{2u_0} e^{-u_0^2}.$$

Since $u_0^2 = \ln(2d)$, we have $e^{-u_0^2} = e^{-\ln(2d)} = \frac{1}{2d}$, hence

$$2d \cdot \frac{L^2}{\sqrt{cn}} \cdot \frac{1}{2u_0} \cdot e^{-u_0^2} = 2d \cdot \frac{L^2}{\sqrt{cn}} \cdot \frac{1}{2u_0} \cdot \frac{1}{2d} = \frac{L^2}{\sqrt{cn}} \cdot \frac{1}{2u_0}.$$

So the tail term is

$$\int_{t_0}^{\infty} 2d \exp\left(-c\,\frac{nt^2}{L^4}\right) dt \le \frac{L^2}{\sqrt{cn}} \cdot \frac{1}{2u_0} = \frac{L^2}{\sqrt{cn}} \cdot \frac{1}{2\sqrt{\ln(2d)}}.$$

Combine and simplify

Combine with $t_0 = \frac{L^2}{\sqrt{cn}}\sqrt{\ln(2d)}$:

$$\mathbb{E}[Z] \le \frac{L^2}{\sqrt{cn}} \left(\sqrt{\ln(2d)} + \frac{1}{2\sqrt{\ln(2d)}}\right).$$

If $d \ge 2$, then $\ln(2d) \ge 1$, so

$$\sqrt{\ln(2d)} + \frac{1}{2\sqrt{\ln(2d)}} \le \sqrt{\ln(2d)} + \frac{1}{2} \le 2\sqrt{\ln(2d)}.$$

Thus

$$\mathbb{E}\|H_n - H\| = \mathbb{E}[Z] \le \frac{2}{\sqrt{c}}\, L^2 \sqrt{\frac{\ln(2d)}{n}} \quad \lesssim \quad L^2 \sqrt{\frac{\log d}{n}}.$$